

LASER X DESIGN

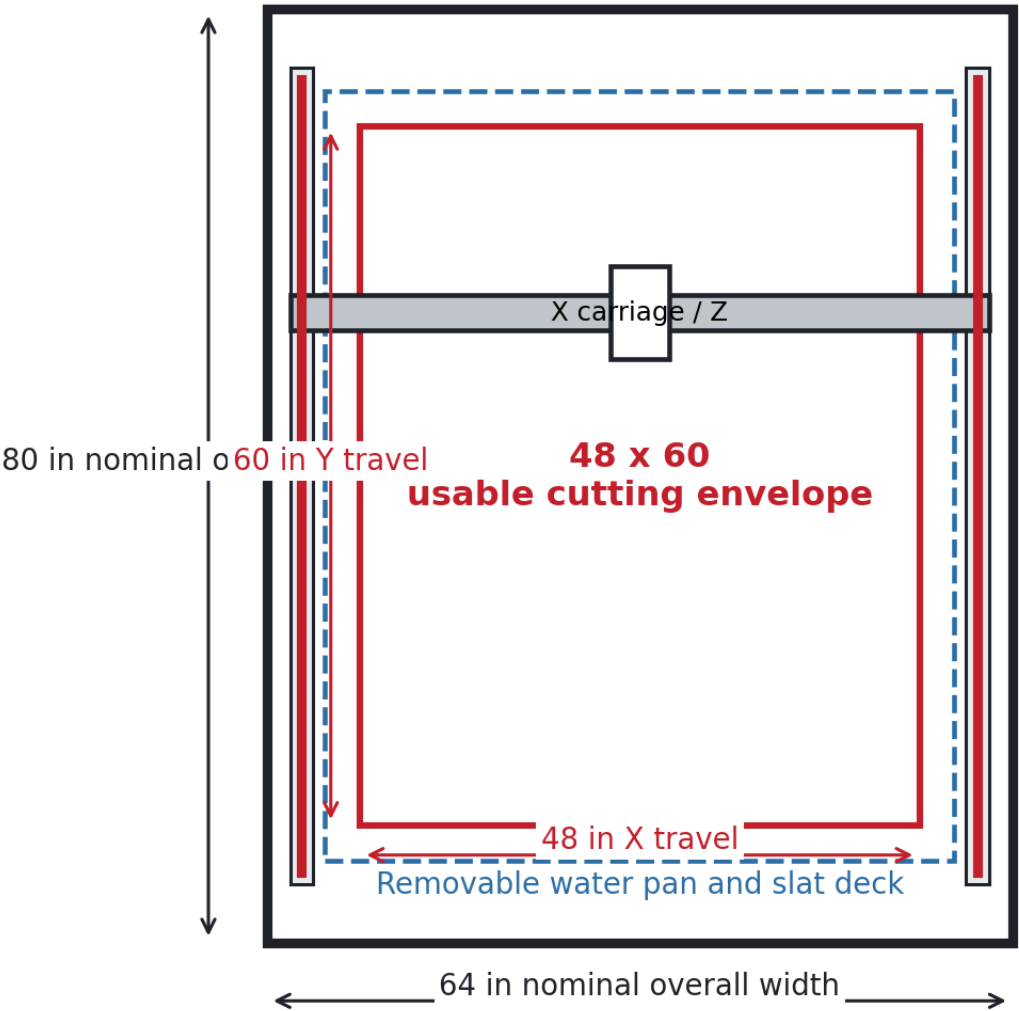
X-1 CNC Plasma Table

Preliminary Engineering & Build Plan

Tube-frame machine | Linear guides | FluidNC control

Revision A - Design Development Release | August 2, 2026

X-1 Preliminary Plan View - Revision A



1. Purpose and Design Status

The X-1 is a low-cost, serviceable CNC plasma table designed around Chris Hilton's existing fabrication capability, free tube and sheet stock, an Everlast PowerPlasma 82i, and a V1 Engineering Jackpot controller running FluidNC.

STATUS

This is a build-planning package, not final production drawings. The frame envelope, guide layout, controls, and assembly process are defined. Exact drive-system hole patterns remain intentionally modular until the X/Y drive is selected.

Design intent

- Build the structural machine primarily from welded 1/8-inch-wall tube already on hand.
- Replace structural 3D-printed bearing blocks and gantry parts with purchased linear guides and laser-cut metal carriages.
- Use laser-cut or formed sheet metal only where it reduces work: motor plates, carriages, guards, water pan, enclosure, and removable cosmetic skirts.
- Target a 48 x 60 inch usable cutting envelope, sized around affordable 1500 mm X and 1800 mm Y linear-motion components.
- Run X, Y1, Y2, and Z under FluidNC, with independent Y home switches for gantry squaring.
- Get the machine cutting at fixed height before adding experimental or custom torch-height control.

What Revision A locks in

Item	Rev A decision
Machine name	X-1 - Laser X Design 1
Construction	Welded tube frame with bolt-on motion components
Guide system	Linear guides on X and Y; compact linear Z
Controller	Existing Jackpot V1.2.1 / ESP32 / FluidNC
Work area	48 x 60 inches nominal
Water control	Separate removable pan and slat deck
Structural printing	None; printing limited to covers, strain reliefs, and mockups
Drive	Modular interface; rack-and-pinion is the baseline recommendation

What remains open

- Final choice between rack-and-pinion, timing belt, or long ball screws for X/Y.
- Final rail family and exact vendor dimensions: SBR20/HGR20 on Y and SBR16/HGR15/HGR20 on X.
- Final motor size and whether the Jackpot drives Z only or feeds external drivers on all axes.
- Final THC architecture.

2. Design Basis and Source Review

The X-1 is an original design informed by two user-supplied reference packages and current controller documentation. The source material is used for design lessons, not copied as a production drawing set.

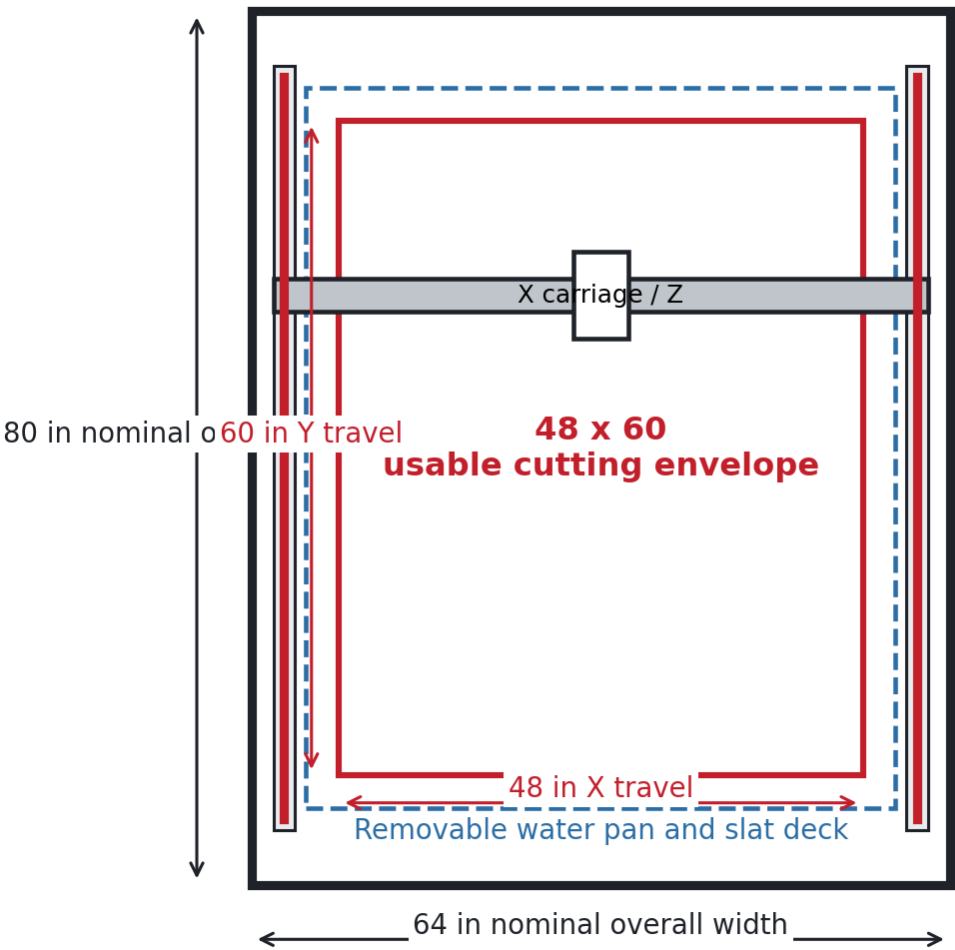
Reference	Useful design lessons carried into X-1
JD's Garage Imperial plan ZIP	Low-cost tube construction, scalable belt-drive concept, NEMA 23 motors, external stepper drivers, ESP32 control, optional Z, and a parts-first build workflow.
CrossFire PRO MAX Assembly Guide	Dual Y drive, separate water table, floating lead nuts, staged rail alignment, bearing preload, gantry tramming, and the requirement to verify parallelism after tightening.
V1 Engineering Jackpot documentation	Six stepper-driver sockets, seven switch inputs, ESP32/FluidNC control, microSD, Wi-Fi/USB, and expansion capability.
FluidNC documentation and examples	Configurable multi-motor axes, standard step/direction output for external drivers, web interface, and independent home-switch concepts for squaring.
Important distinction Linear rails guide the machine. They do not create motion. The X-1 still needs a drive system on each moving axis. Revision A therefore treats guides and drives as separate, replaceable subsystems.	

Source-derived constraints

- The JD package uses 2-inch square tube, belts, NEMA 23 motors, a DM556/TB6600-class driver approach, and structural printed parts that X-1 replaces with metal.
- The CrossFire guide uses dual Y screws and emphasizes assembling the frame loosely, squaring it, moving the gantry through full travel, then completing final tightening.
- The source packages do not provide final dimensions for this 48 x 60 machine; those dimensions are X-1 engineering decisions.

3. Overall Machine Layout

X-1 Preliminary Plan View - Revision A

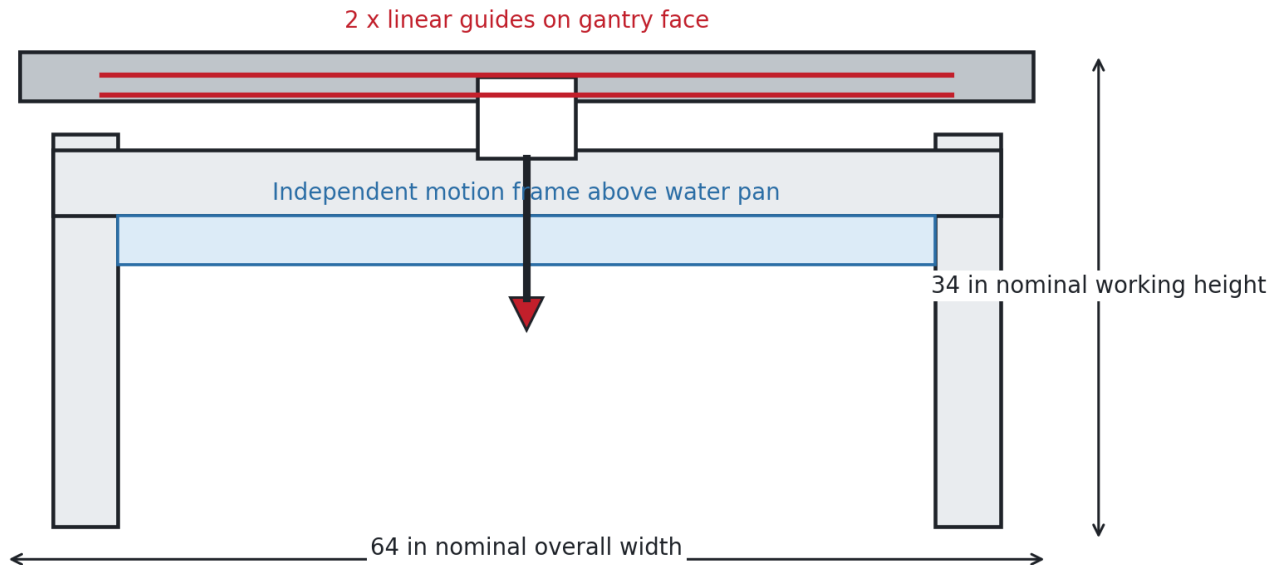


Nominal dimensions; final overall length follows purchased Y guide and drive hardware.

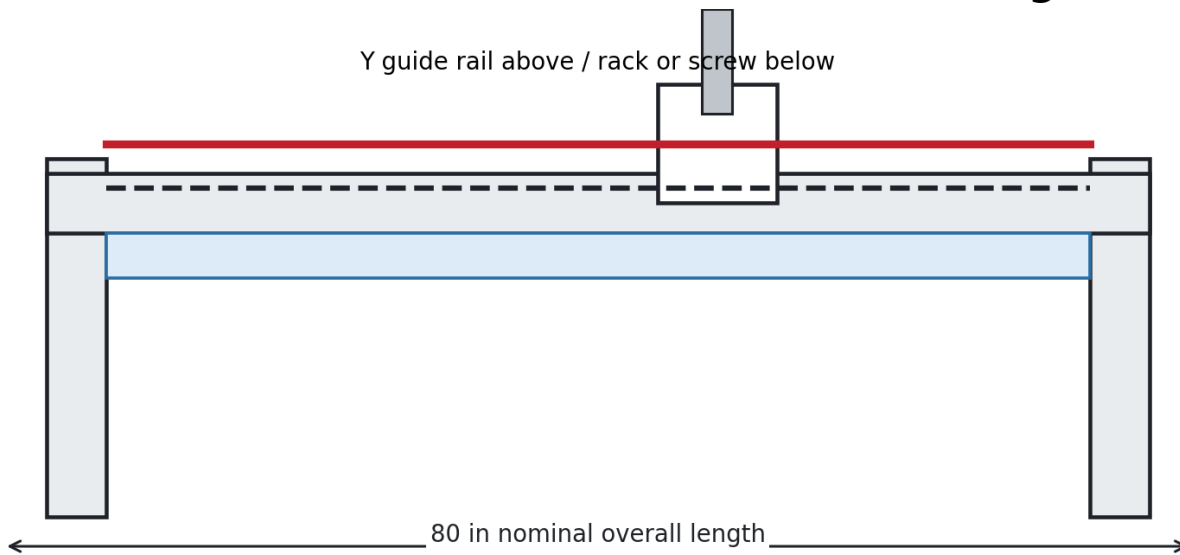
Parameter	Rev A target	Design note
Usable cut width	48 in	Accepts a full 4-foot-wide sheet.
Usable cut length	60 in	Leaves practical travel on an 1800 mm rail package.
Overall width	Approx. 64 in	Includes side beams, rail trucks, and skirt clearance.
Overall length	Approx. 80 in	Includes end hardware and service clearance.
Working height	Approx. 34 in	Adjust after measuring preferred loading height.
Water pan	Approx. 54 x 66 x 3 in	Pan is larger than the cut envelope and isolated from precision rails.
Gantry beam	Approx. 64 in	Actual length set by final truck width and overtravel.
Dimension rule Do not cut rail mounting holes from these nominal dimensions. Purchase or select the exact rails, blocks, rack/screws, end supports, motors, and couplers first; then release Rev B manufacturing drawings from their measured dimensions.		

4. Elevations and Machine Architecture

Front Elevation - Tube Structure with Removable Skirts



Side Elevation - 1800 mm Y Guide Package



Architecture rules

- The rail frame carries only motion loads. The pan and slat supports carry water, material, and dropped parts.
- All cosmetic skirts are removable and mount to rivnuts, weld nuts, or studs; no loose nuts hidden inside tubes.
- Y guides and drives mount on the inside or protected face of the side beams, away from direct slag.
- The gantry is as light as practical. Added weight requires larger motors and reduces acceleration.

5. Structural Frame Weldment

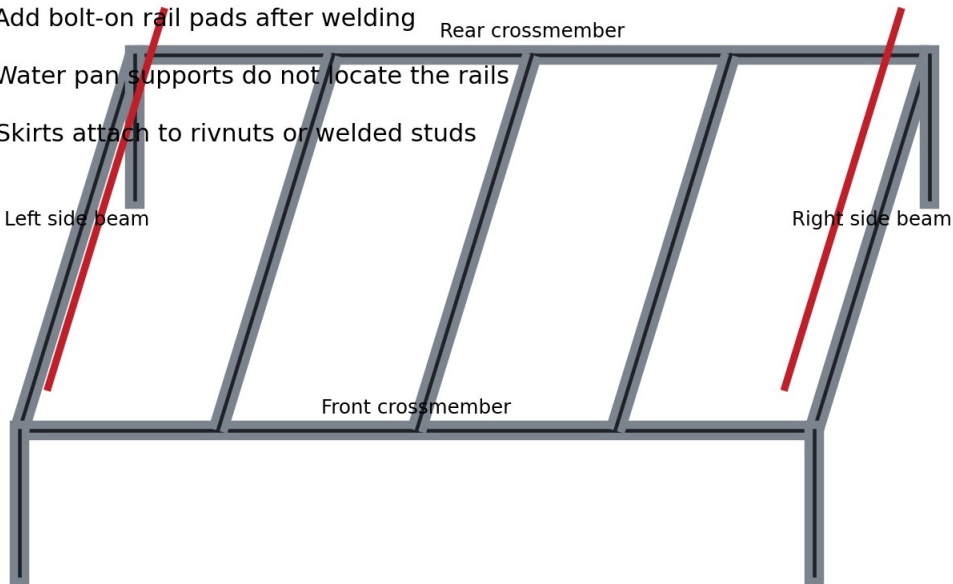
Frame Weldment Concept - Tube First, Sheet Metal Where Useful

A. Weld and square base frame first

B. Add bolt-on rail pads after welding

C. Water pan supports do not locate the rails

D. Skirts attach to rivnuts or welded studs



Provisional tube cut list

Qty	Member	Suggested stock	Nominal length
2	Main Y side beams	2 x 3 x .125 rectangular tube	80 in
3	Main crossmembers	2 x 2 x .125 square tube	58-60 in
4	Legs	2 x 2 x .125 square tube	28-30 in
4	Lower frame braces	2 x 2 x .125 square tube	fit to frame
1	Gantry beam	2 x 3 x .125 rectangular tube	64 in
2	Rail pad strips	1/4 x 3 flat bar or laser-cut plate	72 in
2	Skirt mounting rails	1 x 1 x .083 or tabs	fit to frame

Welding sequence

1. Cut and deburr tube. Verify the two long side beams are the straightest pieces available.
2. Tack the main rectangle on a flat surface. Measure both diagonals and correct until equal.
3. Add center crossmember and leg assemblies with short alternating welds.
4. Allow the frame to cool completely. Recheck twist and diagonal measurements.
5. Do not weld linear rails directly to the frame. Add bolt-on rail pads or drilled/tapped mounting faces after welding.
6. Install pan supports only after rail-beam geometry is stable.

6. Water Pan, Slats, and Skirts

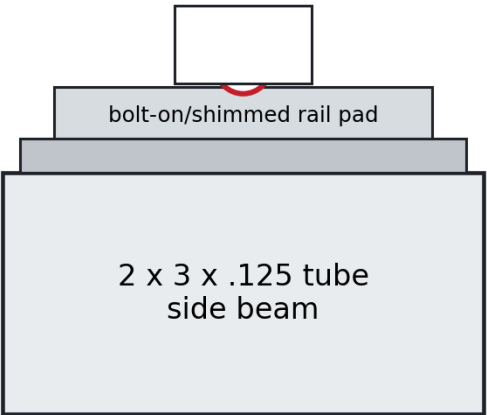
The water pan is a removable consumable system. It must not be used as the precision reference for the motion rails.

Assembly	Construction concept
Water pan	14 or 16 ga formed sheet, 54 x 66 x 3 in nominal. One piece if forming capacity allows; otherwise two flanged halves sealed and bolted like the CrossFire method.
Drain	Two low-point drains or one large drain with a slight formed slope. Use replaceable threaded fittings.
Slat holders	Laser-cut comb strips, bolted to pan ledges. Slats should lift out without removing rails.
Slats	2 to 3 in wide, 11-14 ga strips; slight alternating bow prevents a continuous cut line.
Skirts	Laser-cut cosmetic panels around the tube frame. Mount with rivnuts or welded studs. Panels also shield rack/rails and wiring.
Service access	At least one full-length removable side panel and removable end covers at every motor/drive location.
Keep it easy to live with The X-1 should look like the big laser, but the sheet-metal skins are not structural. They come off quickly so the tube frame, motors, rails, and wiring remain serviceable.	

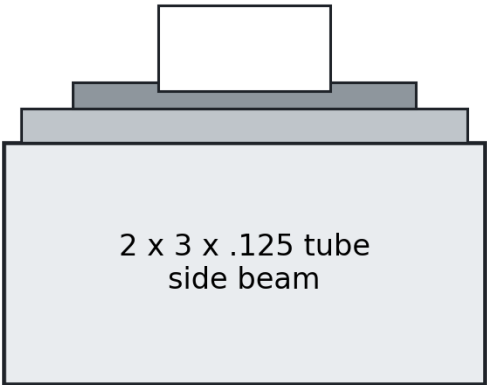
7. Linear Guide System

Guide-Rail Mounting Cross Sections

Option A: SBR20 supported round rail



Option B: HGR20 profile rail



Rail system guides the axis. Drive system creates motion.

Recommended sizes

Axis	Budget choice	Higher-precision choice	Blocks
Y left/right	SBR20, 1800 mm	HGR20, 1800 mm	2 per side
X gantry	Two SBR16, 1500 mm	Two HGR15 or one/two HGR20	2-4 total
Z	MGN15/HGR15, 250-350 mm	Compact purchased Z slide	2 blocks minimum

Why linear guides replace the printed structure

- The purchased blocks carry load through hardened rolling elements instead of printed bearing pockets and spacer stacks.
- Metal carriage plates can be laser cut, bolted, and replaced without reprinting a structural assembly.
- Guide preload and block spacing resist torch-cable twist, gantry pitch, and carriage yaw.
- SBR supported round rails are cheaper and more tolerant of a tube frame; profile rails are stiffer but demand flatter mounting.

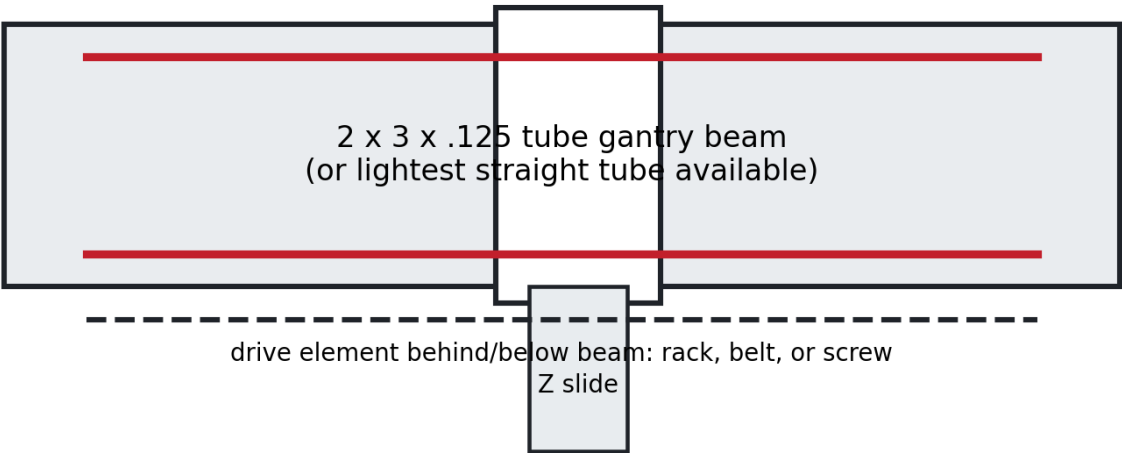
Rail installation target

1. Mount the master Y rail straight and level on shimmed pads.
2. Install both gantry trucks and leave the second rail slightly loose.
3. Move the gantry through full travel and progressively tighten the second rail.
4. Check for consistent push force, no hard spots, and no visible truck rocking.
5. Protect exposed rail surfaces with removable sheet-metal covers and block wipers.

8. Gantry and X Carriage

X-Axis Gantry Concept

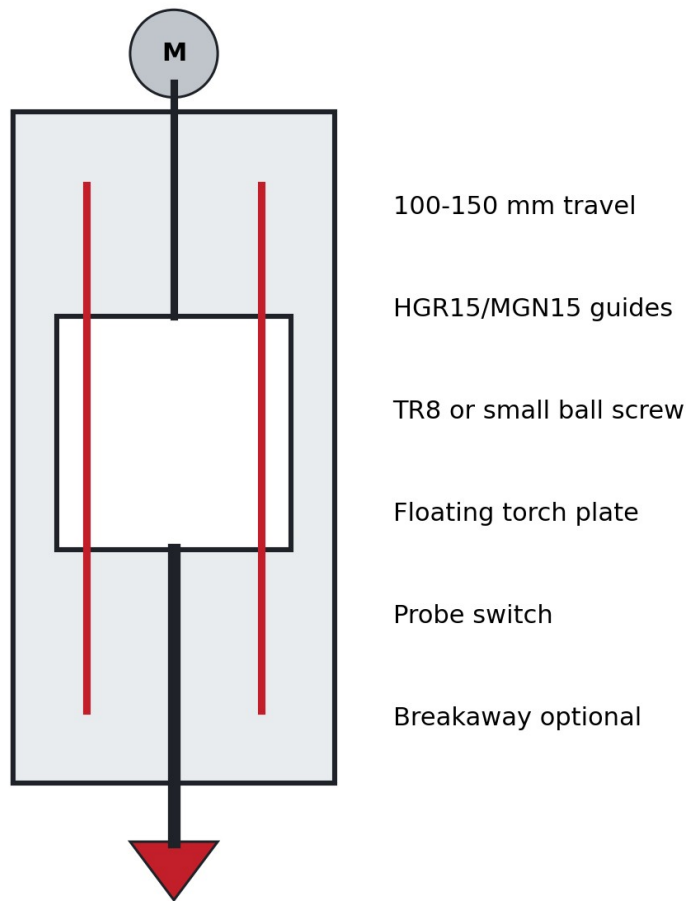
Two rails resist torch cable twist and carriage pitch



Component	Rev A concept
Gantry beam	2 x 3 x .125 tube, 64 in nominal. Use the straightest/lightest available member.
End trucks	Laser-cut 1/4 in plates or compact welded plate boxes. Each Y truck carries two guide blocks and the Y drive interface.
X guides	Two rails on the front face or one profile rail sized to resist moment. Two rails simplify torch-cable torsion.
X carriage	1/4 in laser-cut plate with bolt pattern for Z slide, cable chain, drive nut/pinion, and removable guard.
Serviceability	Motor and idler plates bolt on. No welded motor mount that locks the machine to one drive choice.
Weight target Keep the finished moving gantry assembly under roughly 45 lb if practical. Plasma cutting has almost no cutting force; stiffness and acceleration matter more than massive weight.	

9. Z-Axis and Torch Mount

Z-Axis - Metal Carriage, No Structural Printed Parts



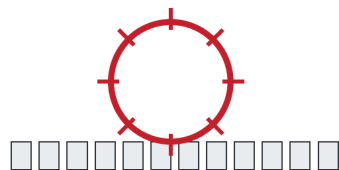
Feature	Requirement
Travel	100-150 mm usable; enough for probing, pierce height, cut height, warped sheet, and service clearance.
Guide	HGR15/MGN15 or purchased compact slide; metal carriage plate.
Drive	TR8 lead screw or small 1204/1605 ball screw. Long-axis speed concerns do not apply on Z.
Motor	NEMA 17 or small NEMA 23; Jackpot onboard TMC driver may remain practical for Z.
Floating head	Separate sliding torch plate with probe switch. Include mechanical hard stop.
Torch holder	Laser-cut/form-and-bolt clamp sized to the machine torch. Avoid printed structural clamps.
Future	Provision for ohmic sensing and breakaway without redesigning the full carriage.

10. X/Y Drive System Decision

Linear Guides Need a Separate Drive System

Rack and pinion RECOMMENDED

Fast, scalable, easy to shield



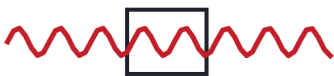
Timing belt LOWEST COST

Cheap and proven, more stretch



Ball screw PRECISION OPTION

Accurate, but long screws cost and whip

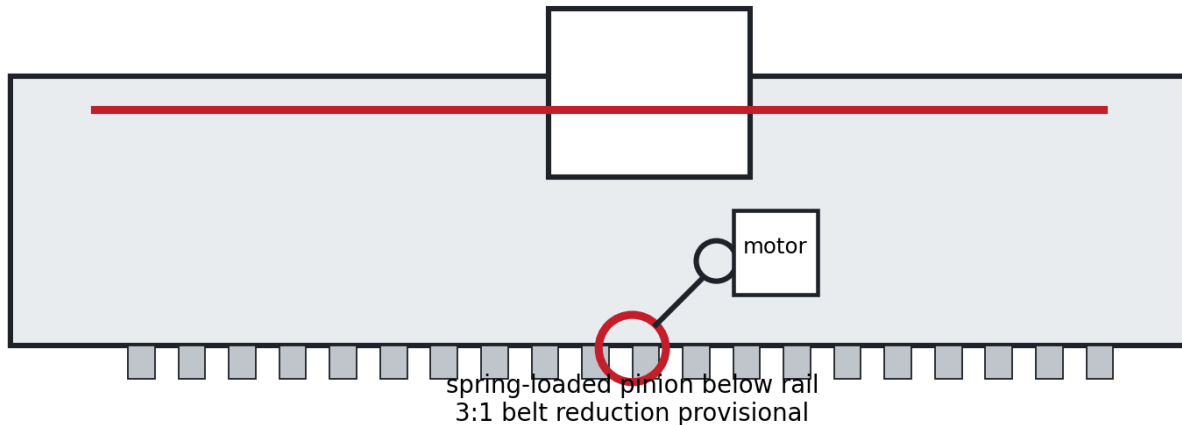


Criterion	Rack & pinion	Timing belt	Ball screw
Long-axis suitability	Excellent	Good	Conditional
Cost	Medium	Lowest	Highest beyond standard lengths
Speed/acceleration	Excellent	Excellent	Limited by screw RPM/whip
Backlash	Low with preload	Low when tensioned	Very low
Contamination tolerance	Good with cover	Fair with cover	Requires excellent protection
Installation on tube frame	Straightforward	Straightforward	Most alignment-sensitive
Future length change	Easy	Easy	Expensive
X-1 recommendation	Baseline	Budget alternate	Precision alternate only
Recommendation Use linear rails for guidance and rack-and-pinion for X, Y1, and Y2. Use a lead screw or ball screw only on Z. This avoids six-foot screw cost and screw-whip risk while keeping the machine fast and scalable.			

11. Baseline Drive - Rack and Pinion

Baseline X-1 Drive Layout - Rack and Pinion

Rack mounts on protected inside/bottom face of tube



Item	Provisional specification
Rack	Module 1 or comparable straight rack, mounted teeth down or inward and covered.
Pinion	20-tooth provisional; final tooth count selected with motor torque and target speed.
Reduction	3:1 timing-belt reduction provisional for torque, resolution, and motor placement.
Y drive	One independent motor/driver per side. Each side gets its own home switch.
X drive	One motor/driver mounted on moving X carriage or gantry end, depending on packaging.
Preload	Spring-loaded or slotted pinion mount to remove lash and accommodate rack joints.
Rack joints	Use standard-length rack sections with a short sacrificial rack section as an alignment gauge during installation.
Protection	Removable skirt/cover over rack and pinion, with access at each motor.

Motor/driver direction

- Use external step/direction drivers for X, Y1, and Y2 if NEMA 23 motors exceed the practical current of the plug-in TMC2209 modules.
- A 36-48 V motor supply improves high-speed torque; the Jackpot itself remains on its rated 9-24 V supply.
- Keep motor-driver adapters and wiring replaceable. Do not modify the Jackpot PCB until the pinout is documented.

12. Alternate Drives

A. Timing belt

- Lowest-cost path and closest to the JD source design.
- Use at least 10-15 mm wide reinforced belt and positive tensioners.
- Protect the belt from sparks and slag. Provide a quick tension check at both ends.
- Best choice if the priority is minimum cash and fastest build rather than maximum industrial feel.

B. Ball screw

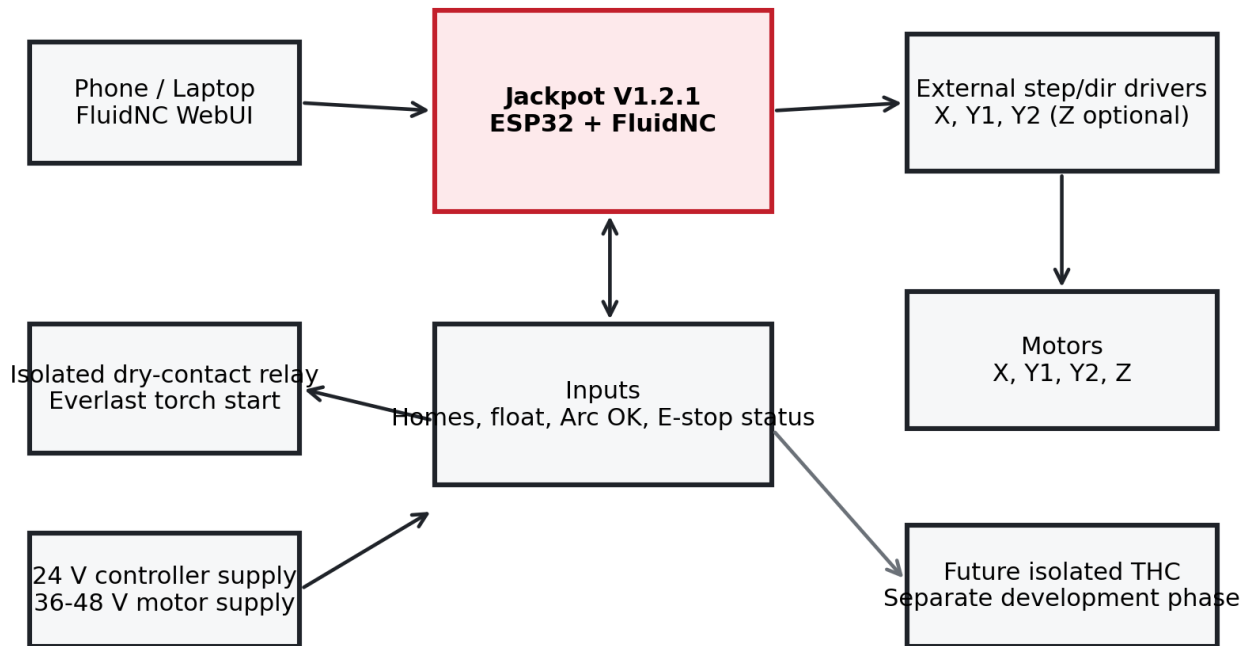
- Do not design the frame around an advertised screw length until the machined end details, nut length, supports, and usable threaded length are known.
- A 72-inch overall screw will not provide 72 inches of travel.
- Prefer 10 mm lead or a fixed-screw/rotating-nut design on a long axis to reduce required screw RPM and whip.
- Use dual independent Y screws and protect them from water, slag, and abrasive dust.
- Ball screws remain attractive if a complete, affordable 1800-2000 mm set becomes available; the X-1 bolt-on motor/end plates can accommodate a later conversion.

Modular interface rule

All drive hardware attaches to replaceable plates. The tube frame and rail geometry stay the same whether the final drive is rack, belt, or screw. This keeps Revision A useful while the drive decision is still being tossed around.

13. FluidNC and Jackpot Control System

X-1 Control Architecture



Function	Planned assignment
X motion	External step/direction driver, or onboard module during bench testing
Y1 motion	Independent motor and home switch
Y2 motion	Independent motor and home switch for squaring
Z motion	Onboard TMC2209 or external driver
Torch command	Jackpot output to isolated dry-contact relay
Arc OK	Everlast signal through appropriate isolation to a Jackpot input
Float probe	Normally closed preferred, filtered input
E-stop	Hardwired removal of motion/torch enable plus status input to controller
THC	Not part of first-motion commissioning; reserved as a separate phase
Controller boundary FluidNC and the Jackpot control motion and basic plasma I/O. They are not a direct analog arc-voltage input. Never connect the Everlast divided-voltage output directly to a Jackpot GPIO input.	

14. Preliminary I/O Map

Signal	Type	Controller assignment	Notes
X motor	Step/Dir	X socket/adaptor	External driver likely
Y1 motor	Step/Dir	Y socket/adaptor	Left side
Y2 motor	Step/Dir	A socket/adaptor	Right side
Z motor	TMC or Step/Dir	Z socket	Onboard driver possible
X home	Digital input	Input 1	NC preferred
Y1 home	Digital input	Input 2	Independent squaring
Y2 home	Digital input	Input 3	Independent squaring
Z home/max	Digital input	Input 4	Optional if float probe is primary
Float probe	Digital input	Input 5	NC, filtered
Arc OK	Isolated digital input	Input 6	Verify Everlast electrical type
E-stop status	Digital input	Input 7	Not the sole safety function
Torch fire	Output to relay	Output 1	Dry contact to plasma start
Fan/water valve	Output	Output 2	Optional
THC voltage	Future isolated interface	Expansion/future controller	No direct connection

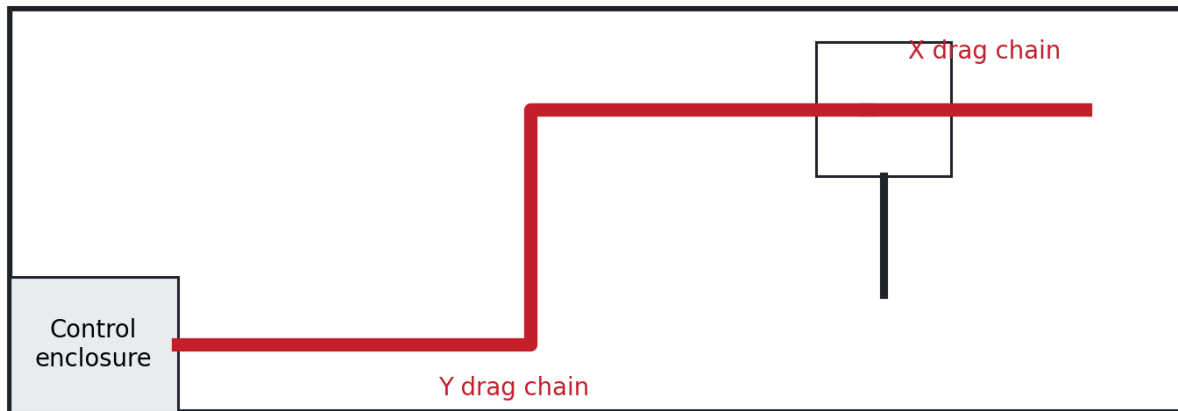
FluidNC configuration deliverables for Rev B

- machine.yaml with axis geometry, motor pins, limits, and homing cycles
- calculated steps/mm for the selected drive
- maximum rates and acceleration established by testing
- startup macro for probing and safe Z position
- torch relay and Arc OK logic
- backed-up configuration in the X-1 GitHub repository

15. Electrical, Grounding, and Cable Routing

Cable Routing and Noise-Control Concept

Keep torch lead separate from motor and switch cables



Shielded motor/signal cable. Ground shields at enclosure end only.

- Mount the electronics enclosure on the frame but electrically manage it as a single control point.
- Route the plasma torch lead separately from stepper, limit, Arc OK, and network cables.
- Use shielded motor and signal cable where practical; terminate shields consistently at the enclosure end.
- Use ferrules, labeled terminals, strain reliefs, and a documented terminal map.
- Keep low-voltage controller power separate from the higher-voltage motor supply.
- Torch start must use an isolated relay and the Everlast interface must be verified before wiring.
- The E-stop must remove the ability to move and fire the torch without relying only on firmware.

Plasma warning

Plasma cutters involve lethal voltage, high-frequency/noisy ignition energy, hot metal, fumes, fire, and automatic motion. Verify the Everlast CNC pinout, isolation, and voltage-divider ratio before any electrical connection.

16. Preliminary Bill of Materials

User-supplied / fabricated

Item	Qty	Notes
1/8-inch-wall structural tube	as required	Main frame, legs, gantry, supports
Flat plate, 1/8-1/4 in	as required	Trucks, motor plates, rack brackets, Z plates
Sheet metal	as required	Pan, guards, skirts, enclosure
Slat material	10-14 pcs	Replaceable
Weld nuts/rivnuts/studs	as required	Removable panels and guards
Everlast PowerPlasma 82i	1	Existing plasma source
Jackpot V1.2.1 controller	1	Existing controller

Purchased motion and control parts

Priority	Item	Qty	Status
1	Y linear rails, 1800 mm, SBR20 or HGR20	2 rails / 4 blocks	Select exact family
1	X linear rails, 1500 mm	1-2 rails / 2-4 blocks	Select exact family
1	Compact Z guide/screw system	1	Select or fabricate
1	X/Y drive components	3 axis sets	Rack baseline; belt/screw alternate
1	NEMA 23 motors	3	Torque selected after gantry estimate
1	External stepper drivers	3-4	DM542/DM556 class or equivalent
1	36-48 V motor power supply	1	Size after motor selection
1	24 V controller power supply	1	Existing may be reused
1	Cable chain	2	Y and X
1	Shielded motor/signal cable	as required	Length after routing
1	Isolated torch relay	1	Dry-contact output
1	Input isolation/filtering	as required	Arc OK and long switch cables
2	Home/limit switches	4-6	Y gets one per side
2	E-stop and contactor/safety relay parts	1 set	Hardwired chain

17. Laser-Cut and Fabricated Parts

Part ID	Part	Material	Qty	Rev A notes
X1-100	Left Y truck plate	1/4 in steel	1	2 guide-block patterns + modular drive plate
X1-101	Right Y truck plate	1/4 in steel	1	Mirrored
X1-110	Gantry end plate	1/4 in steel	2	Bolts tube beam to Y trucks
X1-120	X carriage plate	1/4 in steel	1	Guide blocks, Z, cable chain, drive
X1-130	Z fixed plate	1/4 in aluminum/steel	1	Bolts to X carriage
X1-131	Z floating torch plate	1/8-3/16 in steel	1	Probe travel and hard stop
X1-140	Motor/drive plates	1/4 in steel	3	Release after drive selection
X1-150	Rail pad strips	1/4 in steel	2	Slotted or drilled from measured rails
X1-160	Slat combs	11-14 ga steel	-2	Bolt-in
X1-170	Water pan halves/one-piece	14-16 ga steel	1 set	Drain and flange details
X1-180	Rail/rack guards	16-18 ga steel	4	Removable
X1-190	Exterior skirts	16-18 ga steel	set	Cosmetic, removable, branded
X1-200	Electronics enclosure/backplate	14-16 ga steel	1	Separate clean/dirty wiring zones

Cut-file release

Rev B will contain DXF files and true dimensions after exact guide and drive components are selected and measured. Do not cut motor, rail, or block hole patterns from generic online dimensions.

18. Assembly Sequence

1. Select and purchase the exact X/Y guide components and final drive system.
2. Release Rev B cut list, DXFs, hole patterns, and FluidNC pin map.
3. Cut and weld the main tube frame. Verify diagonals, twist, and leg position.
4. Fabricate and install bolt-on rail pads. Establish the master Y rail.
5. Install Y trucks and second Y rail. Align by moving the gantry through full travel.
6. Build the gantry tube and end trucks. Confirm it sits square without forcing either Y truck.
7. Install X guides, X carriage, and Z assembly. Tram the torch to the slat plane.
8. Install the selected X/Y drives, motors, end supports, tensioners/preload, and guards.
9. Build and install the water pan, slat combs, drains, and removable skirts.
10. Wire power, motors, switches, E-stop status, torch relay, and Arc OK with the plasma disconnected.
11. Configure FluidNC and commission motion at low speed.
12. Verify homing and Y squaring. Then test fixed-height dry runs and finally low-risk plasma cuts.
13. Add THC only after basic motion, probing, torch firing, Arc OK, and cut quality are repeatable.

Hold points

Hold point	Pass condition
HP-1 Frame	Diagonals equal; no rocking; long rail pads can be shimmed straight.
HP-2 Y motion	Gantry moves full length with consistent force and no binding.
HP-3 X/Z	Torch carriage has no perceptible rocking; Z repeats probe position.
HP-4 Electrical	E-stop removes motion/torch capability; no plasma connected during bench test.
HP-5 Dry run	Correct directions, soft limits, homing, squaring, and cable routing.
HP-6 First cut	Fixed-height straight lines before THC or complex CAM.

19. Alignment and Commissioning Checklist

Mechanical

- Level feet and remove frame rock.
- Verify Y master rail straightness and second-rail parallelism.
- Check gantry square after independent Y homing.
- Check X rail parallelism and carriage preload.
- Tram torch front-to-back and side-to-side.
- Confirm rack mesh/belt tension/screw alignment without tight spots.
- Verify hard stops prevent carriage departure from rails.

Electrical

- Verify controller and motor-supply voltages before connection.
- Confirm motor current settings and rotation with plasma unplugged.
- Test every home/probe input by hand in the FluidNC status display.
- Verify E-stop hardware and controller status.
- Verify torch relay dry contact without connecting the Everlast.
- Confirm shield and grounding scheme before routing the torch lead.

Motion tuning

- Start with conservative acceleration and speed.
- Measure commanded versus actual travel over the longest possible distance.
- Tune steps/mm from measured travel, not from pulley or rack calculations alone.
- Increase acceleration until the machine approaches instability, then back off with margin.
- Verify Y1/Y2 squaring after repeated power cycles.

Plasma commissioning

- Connect torch start only after dry testing.
- Test pierce and fixed cut height before any arc-voltage control.
- Run long straight cuts to evaluate mechanical smoothness.
- Keep lead-in/lead-out and CAM issues separate from controller/THC diagnosis.
- Record cut parameters and controller logs for each test.

20. Rev B Decision Sheet

Revision B becomes a true build package with DXFs and finalized dimensions when the following boxes are resolved.

Decision	Options	Recommended / current direction	Final
Y guide	SBR20 / HGR20	SBR20 for budget tolerance; HGR20 for stiffness	[]
X guide	SBR16 / dual HGR15 / HGR20	Dual HGR15 or SBR16	[]
X/Y drive	Rack / belt / ball screw	Rack and pinion	[]
Y guide length	1800 / 2000 mm	1800 mm for about 60 in travel	[]
X guide length	1500 / 1600 mm	1500 mm for 48 in travel	[]
Motor	NEMA 23 torque/current	External-driver NEMA 23	[]
External driver	DM542 / DM556 / existing KL-5056	Reuse existing type if available	[]
Z system	Fabricated / purchased slide	Metal linear slide with screw	[]
Water pan	One piece / two halves	One piece if brake capacity allows	[]
THC	None initially / standalone / custom	Fixed height first; future phase	[]
Next engineering release After the exact guide and drive listings are selected, Rev B should add: final overall dimensions, tube cut lengths, part drawings, DXF cut files, hole tables, rack/screw mounting dimensions, motor brackets, cable lengths, terminal schedule, and a working FluidNC machine.yaml.			

21. Source Notes and Safety

User-supplied source material

- CrossFire PRO MAX Assembly Guide, Langmuir Systems, last updated May 20, 2026, 45 pages.
- JD's Garage Imperial Version plan archive, including parts list, tube/plate drawings, belt components, electronics list, firmware, and optional Z-axis files.
- Photographs of the existing V1 Engineering Jackpot CNC Controller V1.2.1 and TMC2209 modules.
- Screenshot showing SBR12 supported round linear rails as the style reference.

Online technical references checked

- V1 Engineering Jackpot CNC Controller documentation.
- FluidNC project documentation, motor setup examples, and axis-squaring discussions.

Safety limits of this document

This plan does not certify the finished machine, electrical system, plasma interface, fume control, or emergency-stop circuit. The builder is responsible for verifying materials, welds, component ratings, machine guarding, NEC-compliant electrical work, plasma-cutter pinouts, fire protection, ventilation, and safe operating procedures.

Revision A conclusion

The X-1 should be built as a welded tube machine with purchased linear guides and metal carriages. The drive remains a separate decision. Rack-and-pinion is the strongest overall fit for a 48 x 60 FluidNC plasma table, while belt remains the cheapest and ball screw remains an optional precision path if a complete affordable set is found.